Flood situation report — ship_emilia

Event date 2023-05-17 · area 96 km² · generated 2026-10-14T08:22:44Z · sensitivity sensitive

PRELIMINARY — Educational prototype. Results are preliminary, generated automatically from satellite data and not validated in the field.

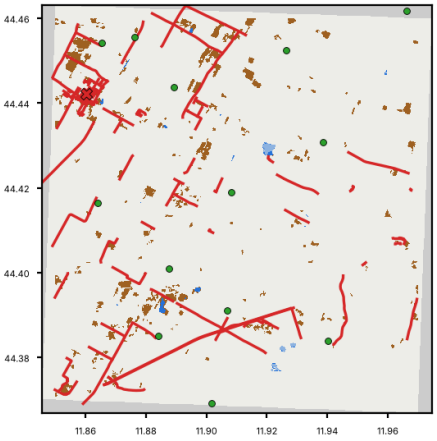
FLOOD DETECTED

3.43 km² affected: 0.13 km² of new flood water and 3.30 km² of debris, mud or other surface change, in the 100 % of the area that the satellites could observe.

Latest image after the event: Radar (Sentinel-1) 2023-05-22 05:19:46 UTC (5 days after the event), track 95 DESCENDING; optical (Sentinel-2) 2023-05-23.

Key numbers

New flood water	**0.13 km²**
Debris, mud, other change	**3.29 km²**
Area affected (total)	**3.43 km²**
of which	**debris 2.53, flooded vegetation 0.27, other change 0.26 km²**
Buildings hit	**232–397 of 12,010 (116 likely destroyed)**
Roads blocked	**74.4 km (+65.0 km possibly) of 521 km**
Bridges blocked / suspect	**1 / 4 of 59**
Settlements cut off	**1 (+0 possibly) of 12**
Long detour (> 2×)	**0**
Important places hit / cut off	**3 / 4 of 38**
Helicopter landing candidates	**3 (unverified; 3 flat)**



Blue = new water, brown = debris/mud, light blue = water before, grey = not observed; red X = cut off, orange = possibly cut off, yellow = long detour, green = connected; red lines = blocked roads.

Most urgent settlements

#	Settlement	Type	Road access	Impact	Bldg hit	Why
1	Sant'Agata sul Santerno	village	cut off	critical	139/623	Cut off: no road connection to a hospital or town
2	Lugo	town	connected	severe	24/4,455	15 buildings likely destroyed (debris or erosion under the foo…
3	Budrio	hamlet	connected	moderate	4/66	2 buildings likely destroyed (debris or erosion under the foot…
4	Chilometro 54	hamlet	connected	moderate	2/35	2 buildings likely destroyed (debris or erosion under the foot…

Repair first

1. Reopen road Via 4 Novembre, Via San Vitale (2 cuts) → reconnects 1 settlement, 623 buildings

Important places hit or cut off

- school: unnamed — likely destroyed, cut off
- school: unnamed — affected, cut off
- airfield: Aeroporto Francesco Baracca — likely destroyed, cut off
- helipad: unnamed — cut off
- helipad: unnamed — possibly cut off

Data quality and main warnings

- Data quality: observed 100 % of the area; radar sees 99 %; optical before/after clear 88 % / 100 %; up to 100 % of the debris area may be false (self-test).
- How sure is the system: overall medium confidence — 13 % of the affected area high, 76 % medium, 11 % low. Main doubts: 100 % of the affected area: the first radar image after the event is 5.2 days late, so water may have receded; 50 % of the affected area is close to the model's decision line (the model is not sure).
- WARNING (debris): This area changes a lot even without a flood: 23.15 km² changed between two earlier radar passes (fields, construction, rain). Similar changes unrelated to the flood may be among the surface changes found.

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